An Official-Source Local RAG System for ShanghaiTech and SIST Question Answering

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Abstract

We build a retrieval-augmented generation (RAG) system for answering questions about ShanghaiTech University and the School of Information Science and Technology (SIST). The system is designed around three constraints: all evidence must come from official sources, all retrieval and generation models must run locally or on self-hosted infrastructure, and evaluation must separate source-evidence quality from final answer correctness. Our implementation collects and cleans 7,190 official documents into 28,481 retrieval chunks, indexes them with SQLite, BM25, and FAISS, and serves cited answers through a Gradio interface backed by a local Qwen-family generator. On a structured 100-question benchmark, the accepted optimized hybrid retriever improves expected source hit@5 from 0.74 to 0.89 over the dense-only before-optimization condition, with higher MRR@5, nDCG@5, and precision@5. For the dense-retrieval generation baseline, a local Qwen3.5 answer run reaches 54/100 automatic accuracy and 57/100 after manual correction of three evaluator false negatives. The optimized generation result remains a placeholder until the matching post-optimization answer run is complete.

1 Introduction

RAG systems are a natural fit for university question answering because many answers depend on authoritative but frequently changing web sources: faculty profiles, course tables, admission notices, program requirements, and event announcements. The goal of this project is to build a local, official-source RAG system that answers ShanghaiTech/SIST questions with inspectable citations rather than relying on a hosted model's parametric memory.

Our system follows a conservative design. Data collection is restricted to official ShanghaiTech and SIST domains. Retrieval artifacts are reproducible from cleaned JSONL records and SQLite metadata. Dense embedding, sparse retrieval, reranking, query expansion, and answer generation use local models or self-hosted runtime paths; no commercial LLM API or hosted inference endpoint is required by the submitted system.

This report makes five contributions:

1. An audited official-source corpus and preprocessing pipeline covering HTML, PDF, Office files, OCR fallback, structured extraction, and append-only crawl records.
2. A complete local RAG pipeline with BM25, FAISS dense retrieval using `BAAI/bge-m3`, optimized hybrid retrieval, local Qwen answer generation, citation validation, and evidence-insufficient fallback.
3. A reviewer-facing Gradio interface that exposes question input, retrieval mode, top-k, retrieval-only inspection, answers, and source contexts.
4. A 100-question retrieval evaluation showing that the optimized hybrid retriever improves expected source evidence quality over the dense-only before-optimization condition.

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5. A dense-retrieval Qwen3.5 generation baseline with answer artifacts, manual review of incorrect outputs, and case studies separating retrieval misses, citation misses, and answer-synthesis failures.

The main optimized-system quantitative claims in this version are retrieval claims. Expected source hit@5, source recall@5, MRR@5, nDCG@5, and precision@5 measure whether the retriever surfaces official evidence, while generated-answer accuracy measures whether the local generator uses that evidence correctly. The dense generation baseline is now reported from real answer artifacts; the after-optimization generation row is intentionally left as a placeholder until the corresponding hybrid answer run and manual review labels are available.

2 Data Collection and Knowledge Base

Official-source scope. The corpus is built from official ShanghaiTech and SIST sources, including the SIST main site, SIST faculty pages, research center subdomains, ShanghaiTech university pages, admissions, graduate admissions, open information, library services, job pages, and academic office pages. Collection runs are append-only: raw crawl outputs remain under `data/collection_runs/`, and accepted outputs are merged into a separate clean dataset under `data/merged/`. This avoids silently overwriting evidence and keeps stale or low-quality pages auditable.

Parsing and cleaning. The crawler extracts HTML text with structured metadata, parses PDFs with text extraction first, uses OCR only as a fallback for low-text PDFs, and treats unsupported binaries as source artifacts rather than retrieval documents. Office files are extracted when local parsers support the format. Chinese and English versions are preserved as distinct documents when both are available. Each document records URL, canonical URL, title, host, category, language, content type, parser, OCR usage, raw path, text path, and content hash.

Table 1: Final clean merged knowledge-base statistics for downstream indexing.

Artifact	Rows
Documents	7,190
Retrieval chunks	28,481
Course records	707
Faculty records	503
Program requirement records	1,777
Event records	5,199

The final clean merged dataset is `data/merged/all-collection-runs-clean-2026-05-27`. JSONL files preserve inspectable records for documents, chunks, courses, faculty members, program requirements, and events. The indexing stage converts these files into a SQLite database for metadata lookup and generated BM25, FAISS, chunk mapping, and build-report artifacts under `data/rag/`.

3 System Architecture

Figure 1 summarizes the implemented system. The same runtime supports retrieval-only inspection, full cited answer generation, Gradio demo use, and structured evaluation.

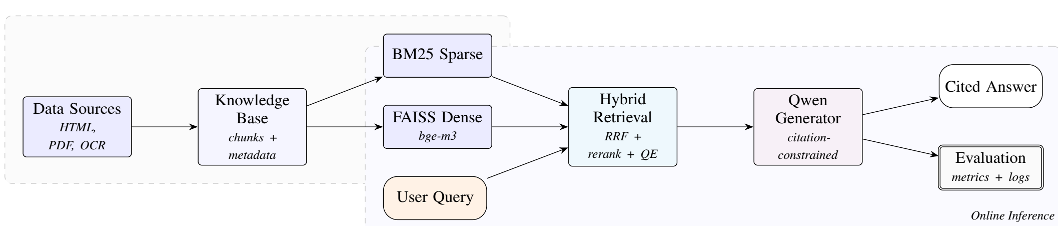

Figure 1: System architecture. Offline indexing builds sparse and dense indexes from official sources. Online inference retrieves, reranks, and generates a citation-constrained answer.

4 Implementation Details

Retrieval components. The sparse retriever uses BM25 over tokenized chunks. Tokenization combines ASCII token matching with `jieba` for Chinese text. The dense retriever embeds chunks and queries with `BAAI/bge-m3`, stores normalized vectors in FAISS, and retrieves by inner product. The hybrid retriever runs sparse and dense candidate searches, merges ranked lists with reciprocal rank fusion (RRF), optionally applies a local CrossEncoder reranker, de-duplicates final sources, and packs the top contexts with title, URL, language, category, snippet, rank, and trace metadata.

Generation components. The answer path is exposed by `rag-answer` and the Python `RagAnswerer`. It supports the official report modes `dense` and `hybrid`. The generator requires an explicit local `-model-path`; model IDs in documentation are acquisition hints, not runtime download defaults. The dense generation baseline in this report was run with a self-hosted Qwen3.5-9B snapshot, `top_k=5`, `max_new_tokens=1024`, `temperature=0.0`, CUDA generation, and the same `BAAI/bge-m3` dense index used by the retrieval baseline. The prompt asks the model to answer in the user's language, use only the provided official-source contexts, preserve label-value bindings, cite factual claims with numbered citations, answer all requested fields when present, and report insufficient evidence when the contexts do not support an answer.

Answer validation and recovery. Generation is not accepted solely because the model produced fluent text. The runtime rejects uncited outputs, citations to missing source IDs, prompt leakage, weak citation support, and missing required contact/profile/credit facts. If the first answer is rejected, the system runs one local repair pass over the same numbered contexts and asks for a strict JSON object with either a cited answer or an explicit insufficient-evidence status. If repair fails, deterministic source-derived fallback covers compact facts such as office/email, address/postcode, course/teacher, dates, credit values, and short list answers. If all three paths fail, the result is recorded as `insufficient_evidence`.

Interface. The Gradio app is implemented as a usable QA interface rather than a landing page. It includes a question box, an ask button, retrieval-mode radio buttons for `hybrid`, `dense`, and `bm25`, a top-k slider, a retrieval-only toggle, a markdown answer panel, source tables, status messages, and sample questions. When no local generator path is configured, the app remains useful as a retrieval-inspection demo.

Evaluation. The evaluation runner `rag-evaluate` consumes `data/test/question_final_structured_100.csv`. The set contains 100 Chinese questions across factual, time-sensitive, comparative, conditional, and multi-hop categories. Each question stores ground-truth answer text, expected official source URLs, evidence snippets, grading notes, and a judge type. Retrieval runs emit JSONL records, summary JSON, review queues, gap notes, and an Excel workbook. Retrieval-only runs intentionally mark answer correctness for manual review because no generated answer is judged.

Generation baseline protocol. The dense generation baseline is `data/eval/generation_dense_qwen35_20260615T052701Z`. It runs `rag-evaluate -runner answer -modes dense` over the same 100-question file and exports JSONL answer records, summary metrics, gap notes, a review queue, an assignment-style Excel workbook, and manual review notes for automatically incorrect answers. The run contains 45 factual, 20 time-sensitive, 13 comparative, 12 multi-hop, and 10 conditional questions. There were no runtime errors, so all reported failures are model, retrieval, citation, or evidence-selection failures rather than crashed examples.

Table 2: Main system components.

Layer	Implementation	Role
Corpus store	JSONL + SQLite	Auditable records and source metadata lookup
Sparse retrieval	BM25 + `jieba`	Exact names, course codes, and Chinese term matching
Dense retrieval	FAISS + `BAAI/bge-m3`	Multilingual semantic retrieval
Hybrid retrieval	RRF, de-duplication, reranking	Optimized evidence selection
Generator	Local Qwen-family model	Cited answer generation from packed contexts
Answer validator	Citation checks, repair prompt, extractive fallback	Reject unsupported answers and recover compact facts
UI	Gradio	Interactive answer and source inspection
Evaluation	`rag-evaluate`	100-question metrics and Excel outputs

5 Retrieval Optimization

The central optimization study compares a dense-only FAISS retriever against a sequence of controlled hybrid-retrieval fixes. The before-optimization baseline embeds the user query with `BAAI/bge-m3`, searches a FAISS index, and returns the top five chunks by dense similarity. It does not use BM25, rank fusion, source de-duplication, enriched index text, reranking, or query expansion. This baseline is strong for semantic matching, but it misses exact course codes, faculty names, URL/path cues, and multi-source questions where several near-duplicate chunks can consume the final context window.

We use the 100-question structured set, the same corpus snapshot, and `top_k=5` throughout. Fix 1 is not counted as a retrieval optimization: it canonicalizes URL and qrels aliases for both conditions so that metrics reflect retrieval behavior rather than harmless source-URL variants. The report-facing baseline is therefore Fix 1 `dense`, rescored under the same canonicalization policy as the final system.

Table 3: Controlled retrieval development path. Rows show the main successful changes after the canonicalized dense baseline. Metrics are retrieval-only evidence metrics.

Condition	Main change relative to prior accepted condition	Hit@1	Hit@5	MRR@5	Latency (s)
Baseline: Fix 1 dense	Dense-only FAISS over `BAAI/bge-m3`; no hybrid fusion or reranking	0.52	0.74	0.594833	2.035649
Fix 1 hybrid control	Add BM25+dense candidates with RRF under shared URL canonicalization	0.56	0.71	0.623167	2.882323
Fix 2 hybrid	Enrich index text with title, category, URL, path tokens, and chunk text	0.64	0.80	0.705833	2.948884
Fix 3 hybrid	Weight dense channel in RRF with sparse weight 1.0 and dense weight 1.5	0.64	0.82	0.710833	2.989514
Fix 4 hybrid	Enforce strict final source diversity to avoid repeated source variants	0.66	0.85	0.735333	2.969989
Fix 6 hybrid	Selectively rerank top-20 candidates while preserving fused top 2	0.65	0.88	0.748333	33.603731
Fix 7 hybrid	Move the same selective reranker to CUDA; ranking unchanged	0.65	0.88	0.748333	5.141354
Fix 8 hybrid	Add conservative local Qwen query expansion before hybrid fusion	0.67	0.89	0.757833	8.389798

From dense baseline to hybrid retrieval. The first optimization axis is adding a sparse channel. Dense retrieval captures semantic similarity, but the evaluation set contains many questions whose evidence depends on exact identifiers: course codes, faculty names, office rooms, admission program names, and URL-specific announcements. Hybrid retrieval adds BM25 candidates and fuses sparse and dense rankings with RRF. The first hybrid control already improves top-rank quality over dense under canonicalized evaluation, raising hit@1 from 0.52 to 0.56 and MRR@5 from 0.594833 to 0.623167. Its hit@5 is lower than dense at this stage, which shows that hybrid fusion alone is not sufficient; the later fixes are needed to recover broad top-five coverage.

Fix 2: enriched index text. The largest single retrieval gain comes from changing the indexed representation rather than the displayed evidence. BM25 tokenization and dense embeddings are built from an enriched text field containing the title, category, canonical URL, URL slug/path tokens, and raw chunk text. Display snippets and packed contexts still use the original official-source text. This helps questions whose answer-bearing page can be identified by page title, category, or URL path even when the body chunk is short or table-like. Hybrid hit@5 rises from 0.71 to 0.80, and MRR@5 rises from 0.623167 to 0.705833.

Fix 3: weighted RRF. After enrichment, dense retrieval remains more reliable than sparse retrieval for many semantic questions. Fix 3 keeps the same candidate pools and index text, but changes hybrid RRF to use `sparse_weight=1.0` and `dense_weight=1.5`. This is a small but clean ranking adjustment: hybrid hit@5 improves from 0.80 to 0.82, MRR@5 improves from 0.705833 to 0.710833, and dense retrieval is unchanged because this fix only affects hybrid fusion.

Fix 4: strict source diversity. Several failures came from repeated source variants occupying multiple top-five slots. Fix 4 caps final output to one item per normalized URL, document ID, and SIST article-ID family. This increases hybrid hit@5 from 0.82/0.83 after rescoring to 0.85 and recovers cases such as q039 and q066 by replacing repeated variants with additional distinct sources. The tradeoff is that nDCG@5 and precision@5 can drop slightly when repeated relevant chunks are replaced by diverse but non-relevant sources. This is still desirable for RAG context packing because a generator benefits from seeing distinct evidence sources.

Fixes 6 and 7: selective local reranking. A full top-50 CrossEncoder reranker was tested but rejected as the default path: it raised hit@5 only to 0.86 while sharply hurting hit@1, MRR@5, nDCG@5, and CPU latency. Fix 6 therefore adds selective reranking: preserve the fused top two candidates and rerank only a bounded top-20 suffix with a local `BAAI/bge-reranker-v2-m3` CrossEncoder. This keeps most first-rank quality while improving coverage: hit@5 rises from 0.85 to 0.88, source recall@5 rises to 0.853333, MRR@5 rises to 0.748333, and nDCG@5 rises to 0.738153. Fix 7 is then a runtime optimization: the same reranker runs on CUDA, preserving the exact ranking while reducing latency from 33.603731 seconds to 5.141354 seconds per query.

Fix 8: conservative query expansion. The final accepted retrieval-ranking change adds one local Qwen3-0.6B rewrite per question. The rewrite is used only as an additional retrieval query; it is not inserted into the answer context and cannot be cited. A direct HyDE-style smoke test hallucinated unsupported offices, emails, and institution names, so the accepted version is deliberately conservative: generated terms are filtered back to the original question vocabulary. This improves hybrid hit@1 from 0.65 to 0.67, hit@5 from 0.88 to 0.89, MRR@5 from 0.748333 to 0.757833, and nDCG@5 from 0.738153 to 0.753315. The cost is latency: Fix 8 averages 8.389798 seconds per query because the expanded query triggers an additional dense candidate pass.

Table 4: Report-facing retrieval-only before/after comparison on the 100-question set. Metrics measure expected official-source evidence quality, not final generated-answer correctness.

Condition	Hit@1	Hit@5	Recall@5	MRR@5	nDCG@5	Precision@5
Before: Fix 1 dense, rescored	0.52	0.74	0.723333	0.594833	0.614946	0.172
After: Fix 8 hybrid expansion	**0.67**	**0.89**	**0.858333**	**0.757833**	**0.753315**	**0.196**
Delta	+0.15	+0.15	+0.135000	+0.163000	+0.138369	+0.024

Overall, the optimized retriever improves every report-facing retrieval metric over the dense baseline. Expected source hit@5 rises from 0.74 to 0.89, meaning that 15 additional questions surface at least one expected official source in the top five retrieved contexts. MRR@5 rises from 0.594833 to 0.757833, showing that expected sources are also ranked earlier, not merely added at the bottom of the context window. The final optimization stack is therefore not a single trick; it combines exact sparse recall, dense semantic recall, richer index text, rank-fusion calibration, source diversity, bounded local reranking, GPU runtime support, and conservative query expansion.

6 Results

6.1 Retrieval Results

Table 4 is the primary quantitative result. The optimized hybrid retriever is stronger than the dense-only before-optimization condition on top-rank quality, top-five coverage, recall, MRR, nDCG, and precision. The result supports a retrieval claim: the system is substantially better at surfacing official evidence after optimization.

Table 5: Top-five expected-source overlap after Fix 8.

Result type	Count	Question IDs
Both hit	81	Not listed individually
Both miss	7	q006, q022, q023, q024, q032, q033, q041
Dense-only hit	4	q003, q013, q018, q050
Hybrid-only hit	8	q029, q031, q037, q039, q046, q066, q072, q099

The remaining failures are not dominated by missing corpus files. For checked disputed questions, expected URLs are often present in the enriched chunk index. The harder cases involve ranking, Chinese/English sibling pages, list pages versus detail pages, faculty profile variants, and course-table or faculty-course join questions that require stronger structured signals than chunk similarity alone.

6.2 Representative Retrieval Cases

Table 6: Representative retrieval-side case studies from the retrieval evaluation. Generation-specific cases are reported separately in Table 8.

Case	Retrieval outcome	Current interpretation
q031	Hybrid-only hit	Selective reranking recovered an expected source that previously ranked below the final top five.
q039	Hybrid-only hit	Strict source diversity helped prevent repeated source variants from consuming context slots.
q046	Hybrid-only hit	Conservative query expansion moved the question from both-miss to hybrid-only hit.
q013	Dense-only hit	Narrowing or reranking can still disturb useful dense evidence for course/faculty join questions.
q022	Both miss	The current retrieval signals remain insufficient; this is a candidate for structured retrieval or qrels audit.

6.3 Dense Generation Baseline

Table 7: Dense-retrieval Qwen3.5 generation baseline. The after-optimization generation row is intentionally left pending.

Metric	Dense generation baseline	Optimized generation
Automatic correct answers	54/100	TODO
Manual-adjusted correct answers	57/100	TODO
Answered-only manual accuracy	57/82 = 69.5%	TODO
Evidence-insufficient answers	18/100	TODO
Remaining wrong or incomplete answers	25/100	TODO
Retrieved expected source hit@5	0.74	TODO
Cited expected source hit@5	0.67	TODO
Answer-synthesis miss rate	0.05	TODO
Average latency	16.095 s	TODO

The dense generation baseline shows that dense retrieval plus a stronger local generator is useful but not sufficient. The run produced 82 answered outputs and 18 evidence-insufficient outputs with no runtime errors. Automatic judging marked 54 answers correct, 28 incorrect, and 18 evidence insufficient. Manual review of the 28 automatically incorrect answers recovered only three clear evaluator false negatives, so the fair baseline score is 57/100 overall rather than a large hidden win masked by the judge. The dominant problem is answer synthesis: 15 of the 28 automatically incorrect answers both retrieved and cited an expected official source, but still omitted required facts or returned an incomplete answer.

Table 8: Representative dense-generation case studies from real answer artifacts.

Case	Query target	Artifact outcome	Analysis
q006	Tu Kewei NLP awards	Correct; retrieved and cited expected source	Dense retrieval finds the faculty/news evidence, and Qwen3.5 extracts the three award facts: ACL 2023 outstanding paper and SemEval 2022/2023 best system paper awards.
q055	Li Quan office, email, and research areas	Correct; retrieved and cited expected source	The system handles a multi-field faculty-profile question when all requested fields are present in one compact profile context.
q004	2025 EE total graduation credits and elective credits	Incorrect despite retrieved/cited expected source	The model cites the correct curriculum pages but says the numeric facts are unavailable, missing 145 total credits and 9 elective credits. This is a table-evidence extraction failure.
q065	PSIT lab directions, quota, email, and PI	Incorrect despite retrieved/cited expected source	The final fallback returns only the contact email and misses research directions, 2–3 annual quota, and Professor Ye Chaofeng. This is a multi-field completeness failure.
q069	2025 CS two-choice course-group rule	Incorrect; expected source not retrieved or cited	Dense retrieval routes the answer to stale 2020 CS and older EE plans, producing 31 elective credits instead of the 2025 rule that overflow credits count toward the 34-credit discipline elective bucket.

These cases motivate the generation-side optimization that is still left as a placeholder in Table 7. The failure artifacts point to table-aware evidence expansion for curriculum pages, multi-field slot planning and validation, stricter year/program routing for degree-plan PDFs, and citation-coverage repair when an expected source is retrieved but not cited. This report does not claim those answer-side optimizations as completed; it records them as the next generation experiment to run against the same 100-question protocol.

7 Discussion and Conclusion

The retrieval study shows that a careful local RAG stack can make large gains without hosted APIs. The final after-optimization retriever combines exact sparse matching, dense semantic matching, rank fusion, source diversity, selective reranking, and conservative query expansion. Together these changes raise expected source hit@5 from 0.74 to 0.89 on the structured 100-question benchmark.

The largest methodological point is that source-evidence metrics must not be confused with answer correctness. A retrieved source can be relevant while the generator still omits a required fact, cites poorly, or returns an evidence-insufficient response. Conversely, a generated answer can look plausible while failing to cite the expected official source. The system therefore keeps retrieval logs, cited sources, answer text, automatic judge status, review queues, and Excel export fields separate.

Limitations. First, only the dense generation baseline has been manually reviewed; the optimized hybrid generation row remains to be inserted after the matching answer run. Second, time-sensitive questions are judged against the corpus snapshot and source dates, so stale official pages can affect both retrieval and answer quality. Third, Fix 8 improves evidence quality but increases latency because expanded queries rerun dense retrieval. Fourth, dense generation failures show that even cited official sources are not enough when table rows, multi-field slots, or year/program constraints are not preserved through answer synthesis.

Future work. The immediate next step is to run the optimized hybrid answer-mode evaluation over the same 100-question set and add manually reviewed after-optimization correctness labels. Answer-side optimization should focus on table-aware evidence expansion for degree plans, multi-field completeness validation, year/program-specific source routing, and citation-coverage repair. Retrieval-side improvements should focus on caching the dense encoder for expanded queries, limiting expansion retrieval to sparse candidates when latency matters, adding contextual chunk enrichment v2 with breadcrumbs and entity aliases, and auditing Chinese/English sibling pages plus answer-bearing faculty/list variants separately from true retrieval changes.

In conclusion, the implemented system satisfies the core project requirements for an official-source, local-model RAG pipeline with an interactive UI and reproducible evaluation artifacts. The current evidence supports a strong retrieval-optimization result and a documented dense generation baseline. Final after-optimization generated-answer claims will be added only after the hybrid answer evaluation produces reviewed correctness labels.

References

[1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*, 2020.

[2] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.

[3] Jeff Johnson, Matthijs Douze, and Herve Jegou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 2019.

[4] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *EMNLP-IJCNLP*, 2019.

[5] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 2009.

[6] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *SIGIR*, 2009.

[7] Qwen Team. Qwen3 technical documentation and model releases. `https://qwenlm.github.io/`.

[8] Gradio Team. Gradio documentation. `https://www.gradio.app/docs`.